Omar Aguilar

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EDUCATION

University of Houston — Houston, TX
Bachelor of Science in Computer Science — *Expected May 2028*

Lone Star College — Houston, TX
Associate of Arts, Computer Science Pathway | GPA: 3.67 — *December 2025*

PROJECT EXPERIENCE

DurableKV | *C++20, Make* — October 2026
Built with AI coding agents · Role: requirements, design approval, testing — *GitHub*

- Log-structured key-value library (put, get, erase, compact) with append-only records protected by CRC32C checksums; recovery truncates a torn final write and refuses to open on mid-log corruption.
- Compaction rewrites live keys to a candidate file, validates it, then swaps it in with rename (Linux) or ReplaceFileW (Windows).
- Test suite covers 254 cases, including seeded randomized runs checked against a std::map model and injected append, flush, index-commit, and compaction-swap failures.

Calendar Intelligence Hub | *TypeScript, React, Node.js, Express, PostgreSQL, Prisma* — October 2026 - Present
Built with AI coding agents · Role: requirements, design approval, workflow testing — *GitHub*

- Full-stack app that ingests Gmail, Outlook, and Discord announcements, extracts events with deterministic rules (no LLM), and routes uncertain dates/times to user confirmation before creating Google Calendar events.
- Duplicate-safe Calendar writes using stable event IDs, payload-hash checks, and database leases; 100 concurrent attempts with restarts produce exactly one insert.
- Covered by 443 Vitest unit, 227 PostgreSQL integration, and 29 Playwright tests.
- Google and Microsoft OAuth tokens stay server-side, encrypted at rest with AES-256-GCM.

PermitWatch | *React, TypeScript, Node.js, Express, SQLite* — October 2026
Best Environment Hack Winner, CodeRED Orion Hackathon, University of Houston — *GitHub*

- Bilingual (English/Spanish) web app for finding environmental permits near a Houston address, built by a team of four.
- Built collectors and normalizers for EPA ECHO wastewater and TCEQ solid-waste data with automated tests, adding 594 permits to grow the Houston-region dataset from 213 to 807 records.
- Extended the haversine permit search with an adjustable 1-10 mile radius that updates the map boundary, markers, and results together.

WORK EXPERIENCE

Digital Personal Shopper — January 2026 - August 2026
Walmart

- Fulfilled 100+ online orders per busy shift, managing substitutions, staging, and customer pickups.

LEADERSHIP EXPERIENCE

CodePathways Peer Mentor — September 2026 - Present
Code[Coogs] x Cougarrettes

- Mentor 1 UH student weekly on coursework, technical skills, and career preparation.

TECHNICAL SKILLS

Languages: C++, C, Python, TypeScript, JavaScript, SQL
Backend & Data: Node.js, Express, REST APIs, OAuth 2.0, PostgreSQL, SQLite, Prisma
Frontend: React, HTML, CSS
Testing & Tools: Vitest, Playwright, Git, Linux, Make